

DSA0108-Object Oriented Programming with C++

Constructor and Destructor

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Q1. Write a C ++ program to create a class for a student with a constructor and a destructor

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", a "C++ PROGRAMMING" tag, and a user profile for "K. Lokesh Chandra" with ID "192425358RD".

On the left, the "QUESTIONS" section shows "5 / 5 completed" and a dropdown menu with "Student" selected. Below this, the "STUDENT" section contains the instruction: "Write a C++ program to create a class for a student with a constructor and a destructor."

The main "PROGRAM EDITOR" area displays the following C++ code:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Student {
5 private:
6     string name;
7     int rollNo;
8 public:
9     Student(string n, int r) {
10         name = n;
11         rollNo = r;
12         cout << "Constructor called: Student object created." << endl;
13     }
14     void display() {
15         cout << "Student Name: " << name << endl;
16         cout << "Roll Number: " << rollNo << endl;
17     }
18     ~Student() {
```

Buttons for "Run" and "Save" are visible. On the right, the "INPUT" field contains the text "Lokesh" and "192425358".

This screenshot shows the same SIMATS Online Programming Lab interface, but with the program executed. The "PROGRAM EDITOR" now shows the final lines of the code:

```
24     s1.display();
25     return 0;
26 }
```

The "OUTPUT" section displays the following text:

```
Constructor called: Student object created.
Student Name: Lokesh
Roll Number: 192425358
Destructor called: Student object destroyed.
```

The "INPUT" field on the right remains empty.

Q2. Write a C++ program to create a class for a book with constructor and a destructor.

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Book

BOOK
Write a C++ program to create a class for a book with a constructor and a destructor.

PROGRAM EDITOR

C++

Run

Save

```
9 public:
10     Book(string t, string a, int p) {
11         title = t;
12         author = a;
13         pages = p;
14         cout << "Book object created!" << endl;
15     }
16     void display() {
17         cout << "Title: " << title << endl;
18         cout << "Author: " << author << endl;
19         cout << "Pages: " << pages << endl;
20     }
21     ~Book() {
22         cout << "Book object destroyed: " << title << endl;
23     }
24 };
25 int main() {
26     Book book1("The C++ Handbook", "Lokesh Chandra", 350);
27     book1.display();
```

INPUT

The C++ Hand Book

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```
26     Book book1("The C++ Handbook", "Lokesh Chandra", 350);
27     book1.display();
28     return 0;
29 }
```

OUTPUT

Book object created!
Title: The C++ Handbook
Author: Lokesh Chandra
Pages: 350
Book object destroyed: The C++ Handbook

Q3. Write a C ++ Program to Create a class for a rectangle with a constructor and a destructor.



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QUESTIONS 5 / 5 completed

Rectangle

RECTANGLE

Write a C++ program to create a class for a rectangle with a constructor and a destructor.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Rectangle {
4 private:
5     float length, width;
6 public:
7     Rectangle(float l, float w) {
8         length = l;
9         width = w;
10        cout << "Constructor called: Rectangle created with length = "
11    }
12    float area() {
13        return length * width;
14    }
15    ~Rectangle() {
16        cout << "Destructor called: Rectangle with length = " << length
17    }
18 };
```

INPUT

10
15



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```
24 cout << "Area of rectangle = " << rect.area() << endl;
25 return 0;
26 }
```

OUTPUT

Enter length and width of rectangle: Constructor called: Rectangle created with length = 0 and width = 9.66896e-44
Area of rectangle = 0
Destructor called: Rectangle with length = 0 and width = 9.66896e-44 destroyed.

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Q4. Write a C++ Program to Create a class for a car with a constructor and destructor.

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QUESTIONS 5 / 5 completed

Car

CAR
Write a C++ program to create a class for a car with a constructor and a destructor.**PROGRAM EDITOR** C++ Run SaveINPUT
toyota

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18 cout << "Car object destroyed: " << brand << endl;
19 }
20 };

21 int main() {

OUTPUT

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Q5. Write a C++ Program to create a class for a bank account with a constructor and destructor.



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Bank

BANK

Write a C++ program to create a class for a bank account with a constructor and a destructor.

```
1 #include <iostream>
2 using namespace std;
3 class BankAccount {
4 private:
5     string accountHolder;
6     long accountNumber;
7     double balance;
8 public:
9     BankAccount(string name, long accNo, double initialBalance) {
10         accountHolder = name;
11         accountNumber = accNo;
12         balance = initialBalance;
13         cout << "BankAccount created for " << accountHolder << endl;
14     }
15     ~BankAccount() {
16         cout << "BankAccount destroyed for " << accountHolder << endl;
17     }
18     void displayDetails() {
19         cout << "\nAccount Holder: " << accountHolder
20             << "\nAccount Number: " << accountNumber
21             << "\nCurrent Balance: " << balance << endl;
```

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```
25 BankAccount acc("Lokesh Chandra", 192425358, 5000.75);
26 acc.displayDetails();
27 return 0;
28 }
```

OUTPUT

```
BankAccount created for Lokesh Chandra

Account Holder: Lokesh Chandra
Account Number: 192425358
Current Balance: 5000.75
BankAccount destroyed for Lokesh Chandra
```

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